

SSS 1-3

FARM-TO-FOOD ACTIVITY SYSTEM

FF-401 to FF-500

SYSTEM -> PRODUCTION -> QUALITY -> PROCESS FLOW -> EFFICIENCY -> SAFETY -> DESIGN

Purpose	Connect school-garden learning to food origin, harvest, handling, quality, preparation pathways, food literacy and waste reduction.
Food-safety boundary	These are education activities, not authorization for unrestricted tasting or processing. School food-safety rules and teacher supervision take priority.
Core learning chain	GARDEN -> CROP -> HARVEST -> HANDLE -> SORT/MEASURE -> PREPARE/PROCESS PATHWAY -> FOOD -> CHOICE -> WASTE/RESOURCE
Connections	Use with GrowMeal practical cards, journals, mission cards, vocabulary and assessment items.

GARDEN 01 - Advanced Production Garden

Ten activities move learners from identifying where food comes from to reasoning about handling, quality, loss, choices and resource recovery.

FF-401 - FOOD ORIGIN

Garden	Advanced Production Garden
Learning objective	Evaluate food origin as part of the Advanced Production Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-402 - HARVEST READINESS

Garden	Advanced Production Garden
Learning objective	Evaluate harvest readiness as part of the Advanced Production Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-403 - SAFE HANDLING

Garden	Advanced Production Garden
Learning objective	Evaluate safe handling as part of the Advanced Production Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-404 - SORTING & QUALITY

Garden	Advanced Production Garden
Learning objective	Evaluate sorting & quality as part of the Advanced Production Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-405 - FARM-TO-FOOD JOURNEY

Garden	Advanced Production Garden
Learning objective	Evaluate farm-to-food journey as part of the Advanced Production Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-406 - MEASUREMENT & YIELD

Garden	Advanced Production Garden
Learning objective	Evaluate measurement & yield as part of the Advanced Production Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-407 - STORAGE & LOSS

Garden	Advanced Production Garden
Learning objective	Evaluate storage & loss as part of the Advanced Production Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-408 - PREPARATION PATHWAY

Garden	Advanced Production Garden
Learning objective	Evaluate preparation pathway as part of the Advanced Production Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-409 - FOOD CHOICE & LITERACY

Garden	Advanced Production Garden
Learning objective	Evaluate food choice & literacy as part of the Advanced Production Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-410 - WASTE-TO-RESOURCE

Garden	Advanced Production Garden
Learning objective	Evaluate waste-to-resource as part of the Advanced Production Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

GARDEN 02 - Crop Trial Garden

Ten activities move learners from identifying where food comes from to reasoning about handling, quality, loss, choices and resource recovery.

FF-411 - FOOD ORIGIN

Garden	Crop Trial Garden
Learning objective	Evaluate food origin as part of the Crop Trial Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-412 - HARVEST READINESS

Garden	Crop Trial Garden
Learning objective	Evaluate harvest readiness as part of the Crop Trial Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-413 - SAFE HANDLING

Garden	Crop Trial Garden
Learning objective	Evaluate safe handling as part of the Crop Trial Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-414 - SORTING & QUALITY

Garden	Crop Trial Garden
Learning objective	Evaluate sorting & quality as part of the Crop Trial Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-415 - FARM-TO-FOOD JOURNEY

Garden	Crop Trial Garden
Learning objective	Evaluate farm-to-food journey as part of the Crop Trial Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-416 - MEASUREMENT & YIELD

Garden	Crop Trial Garden
Learning objective	Evaluate measurement & yield as part of the Crop Trial Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-417 - STORAGE & LOSS

Garden	Crop Trial Garden
Learning objective	Evaluate storage & loss as part of the Crop Trial Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-418 - PREPARATION PATHWAY

Garden	Crop Trial Garden
Learning objective	Evaluate preparation pathway as part of the Crop Trial Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-419 - FOOD CHOICE & LITERACY

Garden	Crop Trial Garden
Learning objective	Evaluate food choice & literacy as part of the Crop Trial Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-420 - WASTE-TO-RESOURCE

Garden	Crop Trial Garden
Learning objective	Evaluate waste-to-resource as part of the Crop Trial Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

GARDEN 03 - Controlled Experiment Garden

Ten activities move learners from identifying where food comes from to reasoning about handling, quality, loss, choices and resource recovery.

FF-421 - FOOD ORIGIN

Garden	Controlled Experiment Garden
Learning objective	Evaluate food origin as part of the Controlled Experiment Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-422 - HARVEST READINESS

Garden	Controlled Experiment Garden
Learning objective	Evaluate harvest readiness as part of the Controlled Experiment Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-423 - SAFE HANDLING

Garden	Controlled Experiment Garden
Learning objective	Evaluate safe handling as part of the Controlled Experiment Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-424 - SORTING & QUALITY

Garden	Controlled Experiment Garden
Learning objective	Evaluate sorting & quality as part of the Controlled Experiment Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-425 - FARM-TO-FOOD JOURNEY

Garden	Controlled Experiment Garden
Learning objective	Evaluate farm-to-food journey as part of the Controlled Experiment Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-426 - MEASUREMENT & YIELD

Garden	Controlled Experiment Garden
Learning objective	Evaluate measurement & yield as part of the Controlled Experiment Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-427 - STORAGE & LOSS

Garden	Controlled Experiment Garden
Learning objective	Evaluate storage & loss as part of the Controlled Experiment Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-428 - PREPARATION PATHWAY

Garden	Controlled Experiment Garden
Learning objective	Evaluate preparation pathway as part of the Controlled Experiment Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-429 - FOOD CHOICE & LITERACY

Garden	Controlled Experiment Garden
Learning objective	Evaluate food choice & literacy as part of the Controlled Experiment Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-430 - WASTE-TO-RESOURCE

Garden	Controlled Experiment Garden
Learning objective	Evaluate waste-to-resource as part of the Controlled Experiment Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

GARDEN 04 - Irrigation Engineering Garden

Ten activities move learners from identifying where food comes from to reasoning about handling, quality, loss, choices and resource recovery.

FF-431 - FOOD ORIGIN

Garden	Irrigation Engineering Garden
Learning objective	Evaluate food origin as part of the Irrigation Engineering Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-432 - HARVEST READINESS

Garden	Irrigation Engineering Garden
Learning objective	Evaluate harvest readiness as part of the Irrigation Engineering Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-433 - SAFE HANDLING

Garden	Irrigation Engineering Garden
Learning objective	Evaluate safe handling as part of the Irrigation Engineering Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-434 - SORTING & QUALITY

Garden	Irrigation Engineering Garden
Learning objective	Evaluate sorting & quality as part of the Irrigation Engineering Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-435 - FARM-TO-FOOD JOURNEY

Garden	Irrigation Engineering Garden
Learning objective	Evaluate farm-to-food journey as part of the Irrigation Engineering Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-436 - MEASUREMENT & YIELD

Garden	Irrigation Engineering Garden
Learning objective	Evaluate measurement & yield as part of the Irrigation Engineering Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-437 - STORAGE & LOSS

Garden	Irrigation Engineering Garden
Learning objective	Evaluate storage & loss as part of the Irrigation Engineering Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-438 - PREPARATION PATHWAY

Garden	Irrigation Engineering Garden
Learning objective	Evaluate preparation pathway as part of the Irrigation Engineering Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-439 - FOOD CHOICE & LITERACY

Garden	Irrigation Engineering Garden
Learning objective	Evaluate food choice & literacy as part of the Irrigation Engineering Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-440 - WASTE-TO-RESOURCE

Garden	Irrigation Engineering Garden
Learning objective	Evaluate waste-to-resource as part of the Irrigation Engineering Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

GARDEN 05 - Water Productivity Garden

Ten activities move learners from identifying where food comes from to reasoning about handling, quality, loss, choices and resource recovery.

FF-441 - FOOD ORIGIN

Garden	Water Productivity Garden
Learning objective	Evaluate food origin as part of the Water Productivity Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-442 - HARVEST READINESS

Garden	Water Productivity Garden
Learning objective	Evaluate harvest readiness as part of the Water Productivity Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-443 - SAFE HANDLING

Garden	Water Productivity Garden
Learning objective	Evaluate safe handling as part of the Water Productivity Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-444 - SORTING & QUALITY

Garden	Water Productivity Garden
Learning objective	Evaluate sorting & quality as part of the Water Productivity Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-445 - FARM-TO-FOOD JOURNEY

Garden	Water Productivity Garden
Learning objective	Evaluate farm-to-food journey as part of the Water Productivity Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-446 - MEASUREMENT & YIELD

Garden	Water Productivity Garden
Learning objective	Evaluate measurement & yield as part of the Water Productivity Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-447 - STORAGE & LOSS

Garden	Water Productivity Garden
Learning objective	Evaluate storage & loss as part of the Water Productivity Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-448 - PREPARATION PATHWAY

Garden	Water Productivity Garden
Learning objective	Evaluate preparation pathway as part of the Water Productivity Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-449 - FOOD CHOICE & LITERACY

Garden	Water Productivity Garden
Learning objective	Evaluate food choice & literacy as part of the Water Productivity Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-450 - WASTE-TO-RESOURCE

Garden	Water Productivity Garden
Learning objective	Evaluate waste-to-resource as part of the Water Productivity Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

GARDEN 06 - Advanced Soilless Garden

Ten activities move learners from identifying where food comes from to reasoning about handling, quality, loss, choices and resource recovery.

FF-451 - FOOD ORIGIN

Garden	Advanced Soiless Garden
Learning objective	Evaluate food origin as part of the Advanced Soiless Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-452 - HARVEST READINESS

Garden	Advanced Soilless Garden
Learning objective	Evaluate harvest readiness as part of the Advanced Soilless Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-453 - SAFE HANDLING

Garden	Advanced Soilless Garden
Learning objective	Evaluate safe handling as part of the Advanced Soilless Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-454 - SORTING & QUALITY

Garden	Advanced Soilless Garden
Learning objective	Evaluate sorting & quality as part of the Advanced Soilless Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-455 - FARM-TO-FOOD JOURNEY

Garden	Advanced Soilless Garden
Learning objective	Evaluate farm-to-food journey as part of the Advanced Soilless Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-456 - MEASUREMENT & YIELD

Garden	Advanced Soilless Garden
Learning objective	Evaluate measurement & yield as part of the Advanced Soilless Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-457 - STORAGE & LOSS

Garden	Advanced Soilless Garden
Learning objective	Evaluate storage & loss as part of the Advanced Soilless Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-458 - PREPARATION PATHWAY

Garden	Advanced Soilless Garden
Learning objective	Evaluate preparation pathway as part of the Advanced Soilless Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-459 - FOOD CHOICE & LITERACY

Garden	Advanced Soiless Garden
Learning objective	Evaluate food choice & literacy as part of the Advanced Soiless Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-460 - WASTE-TO-RESOURCE

Garden	Advanced Soilless Garden
Learning objective	Evaluate waste-to-resource as part of the Advanced Soilless Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

GARDEN 07 - Sensor & Monitoring Garden

Ten activities move learners from identifying where food comes from to reasoning about handling, quality, loss, choices and resource recovery.

FF-461 - FOOD ORIGIN

Garden	Sensor & Monitoring Garden
Learning objective	Evaluate food origin as part of the Sensor & Monitoring Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-462 - HARVEST READINESS

Garden	Sensor & Monitoring Garden
Learning objective	Evaluate harvest readiness as part of the Sensor & Monitoring Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-463 - SAFE HANDLING

Garden	Sensor & Monitoring Garden
Learning objective	Evaluate safe handling as part of the Sensor & Monitoring Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-464 - SORTING & QUALITY

Garden	Sensor & Monitoring Garden
Learning objective	Evaluate sorting & quality as part of the Sensor & Monitoring Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-465 - FARM-TO-FOOD JOURNEY

Garden	Sensor & Monitoring Garden
Learning objective	Evaluate farm-to-food journey as part of the Sensor & Monitoring Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-466 - MEASUREMENT & YIELD

Garden	Sensor & Monitoring Garden
Learning objective	Evaluate measurement & yield as part of the Sensor & Monitoring Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-467 - STORAGE & LOSS

Garden	Sensor & Monitoring Garden
Learning objective	Evaluate storage & loss as part of the Sensor & Monitoring Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-468 - PREPARATION PATHWAY

Garden	Sensor & Monitoring Garden
Learning objective	Evaluate preparation pathway as part of the Sensor & Monitoring Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-469 - FOOD CHOICE & LITERACY

Garden	Sensor & Monitoring Garden
Learning objective	Evaluate food choice & literacy as part of the Sensor & Monitoring Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-470 - WASTE-TO-RESOURCE

Garden	Sensor & Monitoring Garden
Learning objective	Evaluate waste-to-resource as part of the Sensor & Monitoring Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

GARDEN 08 - Smart Irrigation Demonstrator

Ten activities move learners from identifying where food comes from to reasoning about handling, quality, loss, choices and resource recovery.

FF-471 - FOOD ORIGIN

Garden	Smart Irrigation Demonstrator
Learning objective	Evaluate food origin as part of the Smart Irrigation Demonstrator food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-472 - HARVEST READINESS

Garden	Smart Irrigation Demonstrator
Learning objective	Evaluate harvest readiness as part of the Smart Irrigation Demonstrator food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-473 - SAFE HANDLING

Garden	Smart Irrigation Demonstrator
Learning objective	Evaluate safe handling as part of the Smart Irrigation Demonstrator food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-474 - SORTING & QUALITY

Garden	Smart Irrigation Demonstrator
Learning objective	Evaluate sorting & quality as part of the Smart Irrigation Demonstrator food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-475 - FARM-TO-FOOD JOURNEY

Garden	Smart Irrigation Demonstrator
Learning objective	Evaluate farm-to-food journey as part of the Smart Irrigation Demonstrator food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-476 - MEASUREMENT & YIELD

Garden	Smart Irrigation Demonstrator
Learning objective	Evaluate measurement & yield as part of the Smart Irrigation Demonstrator food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-477 - STORAGE & LOSS

Garden	Smart Irrigation Demonstrator
Learning objective	Evaluate storage & loss as part of the Smart Irrigation Demonstrator food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-478 - PREPARATION PATHWAY

Garden	Smart Irrigation Demonstrator
Learning objective	Evaluate preparation pathway as part of the Smart Irrigation Demonstrator food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-479 - FOOD CHOICE & LITERACY

Garden	Smart Irrigation Demonstrator
Learning objective	Evaluate food choice & literacy as part of the Smart Irrigation Demonstrator food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-480 - WASTE-TO-RESOURCE

Garden	Smart Irrigation Demonstrator
Learning objective	Evaluate waste-to-resource as part of the Smart Irrigation Demonstrator food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

GARDEN 09 - Research & Design Garden

Ten activities move learners from identifying where food comes from to reasoning about handling, quality, loss, choices and resource recovery.

FF-481 - FOOD ORIGIN

Garden	Research & Design Garden
Learning objective	Evaluate food origin as part of the Research & Design Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-482 - HARVEST READINESS

Garden	Research & Design Garden
Learning objective	Evaluate harvest readiness as part of the Research & Design Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-483 - SAFE HANDLING

Garden	Research & Design Garden
Learning objective	Evaluate safe handling as part of the Research & Design Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-484 - SORTING & QUALITY

Garden	Research & Design Garden
Learning objective	Evaluate sorting & quality as part of the Research & Design Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-485 - FARM-TO-FOOD JOURNEY

Garden	Research & Design Garden
Learning objective	Evaluate farm-to-food journey as part of the Research & Design Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-486 - MEASUREMENT & YIELD

Garden	Research & Design Garden
Learning objective	Evaluate measurement & yield as part of the Research & Design Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-487 - STORAGE & LOSS

Garden	Research & Design Garden
Learning objective	Evaluate storage & loss as part of the Research & Design Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-488 - PREPARATION PATHWAY

Garden	Research & Design Garden
Learning objective	Evaluate preparation pathway as part of the Research & Design Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-489 - FOOD CHOICE & LITERACY

Garden	Research & Design Garden
Learning objective	Evaluate food choice & literacy as part of the Research & Design Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-490 - WASTE-TO-RESOURCE

Garden	Research & Design Garden
Learning objective	Evaluate waste-to-resource as part of the Research & Design Garden food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

GARDEN 10 - Integrated Food Systems Laboratory

Ten activities move learners from identifying where food comes from to reasoning about handling, quality, loss, choices and resource recovery.

FF-491 - FOOD ORIGIN

Garden	Integrated Food Systems Laboratory
Learning objective	Evaluate food origin as part of the Integrated Food Systems Laboratory food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-492 - HARVEST READINESS

Garden	Integrated Food Systems Laboratory
Learning objective	Evaluate harvest readiness as part of the Integrated Food Systems Laboratory food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-493 - SAFE HANDLING

Garden	Integrated Food Systems Laboratory
Learning objective	Evaluate safe handling as part of the Integrated Food Systems Laboratory food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-494 - SORTING & QUALITY

Garden	Integrated Food Systems Laboratory
Learning objective	Evaluate sorting & quality as part of the Integrated Food Systems Laboratory food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-495 - FARM-TO-FOOD JOURNEY

Garden	Integrated Food Systems Laboratory
Learning objective	Evaluate farm-to-food journey as part of the Integrated Food Systems Laboratory food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-496 - MEASUREMENT & YIELD

Garden	Integrated Food Systems Laboratory
Learning objective	Evaluate measurement & yield as part of the Integrated Food Systems Laboratory food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-497 - STORAGE & LOSS

Garden	Integrated Food Systems Laboratory
Learning objective	Evaluate storage & loss as part of the Integrated Food Systems Laboratory food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-498 - PREPARATION PATHWAY

Garden	Integrated Food Systems Laboratory
Learning objective	Evaluate preparation pathway as part of the Integrated Food Systems Laboratory food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-499 - FOOD CHOICE & LITERACY

Garden	Integrated Food Systems Laboratory
Learning objective	Evaluate food choice & literacy as part of the Integrated Food Systems Laboratory food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-500 - WASTE-TO-RESOURCE

Garden	Integrated Food Systems Laboratory
Learning objective	Evaluate waste-to-resource as part of the Integrated Food Systems Laboratory food-system chain, integrating production, quality, process efficiency, safety constraints and design decisions.
Materials	Research journal, system/process map, measurement instruments, quality records, calculator/computer where available, clean handling materials.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define system boundary and performance objective. 2. Map the product/material flow. 3. Quantify a production, quality, loss, storage or process metric. 4. Identify bottleneck and uncertainty. 5. Compare two management/design scenarios. 6. Recommend an optimization with acceptance criteria.
Evidence to record	Process-flow diagram, quantitative metric, scenario comparison, uncertainty note and defensible recommendation.
Farm-to-Food challenge	Propose a next-stage test or prototype that improves farm-to-food performance without weakening safety.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?